

Brian Zinkan

AI-Native Full-Stack Builder | Founding Engineer

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SUMMARY

AI-native builder who ships production SaaS at vibecoding speed. Solo-built five full-stack platforms across EdTech, regulatory AI, and citizen science using AI-first workflows — from multi-agent LLM orchestration (CrewAI + Inngest) to real-time classroom infrastructure on AWS and GCP. Comfortable owning the entire stack: schema design, backend, frontend, AI pipelines, cloud infrastructure, security programs, and compliance.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Frontend: React, Next.js 14, Vite, Tailwind CSS, Radix/shadcn, Capacitor (iOS/Android), Expo (React Native)

Backend: FastAPI, Express, Node.js, Drizzle ORM, Prisma, SQLAlchemy, Pydantic, Mangum (Lambda)

Data / Real-Time: PostgreSQL 16, pgvector (HNSW), PostGIS, DynamoDB, Redis, Socket.io, WebSocket, WebRTC

Orchestration: Inngest (durable execution, step functions, fan-out), CrewAI (multi-agent), cron scheduling

AI / ML: Claude, Gemini 2.5, OpenAI, AWS Bedrock, Veo 3, RAG pipelines, multi-LLM routing

AWS: ECS/Fargate, RDS, S3, CloudFront, ECR, SQS, SES, Lambda, API Gateway, Bedrock, Transcribe, Cognito, Polly

GCP: Cloud Run, Cloud SQL, Cloud Storage, Eventarc, Cloud Tasks, Cloud Scheduler, Secret Manager, Firebase Auth, Cloud Build, Artifact Registry, Cloud DNS

Infra / CI: Docker, docker-compose, GitHub Actions, multi-stage Dockerfiles

Auth / Pay: NextAuth + JWT, Passport, Google OAuth, Cognito JWT, Stripe (subscriptions + credits), SendGrid, Resend

Security: WISP, HECVAT Lite, FERPA/COPPA, audit logging, security monitoring, AES-256, TLS 1.2+

PROJECTS

Lulia Lesson Lab

AI-powered lesson generation platform | Python, FastAPI, Next.js, PostgreSQL + pgvector, AWS, Inngest, CrewAI

- Architected a 16-agent, 5-crew AI orchestration system (CrewAI) that auto-generates standards-aligned lessons, worksheets, activities, games, and narrated videos from uploaded curriculum.
- Inngest durable execution: 8 functions, 4 event-driven workflows (plan approval, video upload with 8 checkpointed steps, Veo 3 clips, video crew pipeline), 3 cron jobs. Step-level memoization, concurrency limits, automatic retries.
- Three-tier standards router with dense retrieval over 1.21M education standards in pgvector (HNSW) using Bedrock embeddings. Multi-LLM routing: Claude, Gemini 2.5 Flash, Haiku, Bedrock — fallback policies on every call.
- SHA-256 fingerprinting prevents content repetition. Accommodation engine supports IEP, 504, ELL, and Gifted learners.
- Production: Docker, ECS/Fargate, RDS, S3, SQS, SES, CloudFront, Bedrock, Transcribe. GitHub Actions CI/CD.

SchoolPilot (ClassPilot + PassPilot + GoPilot)

Unified K-12 SaaS platform | TypeScript, React, Express, PostgreSQL, Redis, Socket.io, WebRTC, Capacitor, AWS

- Built three standalone K-12 products — classroom monitoring (WebRTC), digital hall passes (Capacitor mobile), student dismissal (real-time bus/car-line) — then unified under one tenant-aware SaaS with shared auth and tiered pricing.
- Dual real-time channels: Socket.io for dismissal events, raw WebSocket for device streaming. Enterprise security: WISP, HECVAT Lite, real-time monitor (deterministic rules, 5-min cycles), audit logging with 2-year retention.
- FERPA/COPPA compliance. Dual JWT + session cookies, RBAC (6 roles), account lockout, IP rate limiting, AES-GCM token encryption, MFA.
- AWS: ECS/Fargate, ECR, CloudFront, ALB, RDS, S3. Versioned releases. GitHub Actions CI/CD.

Gordi

Regulatory compliance decision engine | TypeScript, Next.js 14, Prisma, PostgreSQL, PostGIS, AWS Fargate, Inngest

- Hybrid AI + deterministic rules engine that turns fragmented local ordinances into defensible compliance answers with legal citations. Deterministic layer (PROHIBITED > RESTRICTED > CONDITIONAL > ALLOWED) ensures reproducible, auditable results.
- Multi-jurisdiction coverage (Denver, Boston, DC, NYC, Seattle). Real-time ordinance monitoring tracks new laws and alerts businesses. PostGIS + S3 polygon storage for geospatial zoning/parcel lookup.
- Inngest for async workflows. Upstash Redis rate limiting. Monetization: free tier + paid compliance reports (\$9.99–\$49.99) + permit workflows. Stripe, Resend, UTM + referral analytics.
- Multi-provider AI: Claude, Gemini, Bedrock. Dockerized on AWS Fargate with ECR, S3, CloudWatch.

Dragonfly

Citizen science field app for kids | Python, FastAPI, Expo (React Native), Cloud SQL, GCP Cloud Run, Firebase Auth

- Mobile-first field app for ages 9–12 that turns nature observations into iNaturalist submissions, fills a personal Pokédex-style "Dex," claims map territory, and drives class/friend-group leaderboards.
- GCP-native architecture: Cloud Run (FastAPI + uvicorn), Cloud SQL for PostgreSQL, Cloud Storage (signed URLs, lifecycle rules), Artifact Registry, Cloud Build, Secret Manager, Cloud DNS.
- Async pipelines: Eventarc (GCS triggers for photo moderation), Cloud Tasks (iNaturalist submission queue with DLQ + retry), Cloud Scheduler (nightly rarity refresh cron).
- Firebase Auth (custom claims for parent/teacher/kid roles, join-code auth for under-13). Expo (React Native) for iOS/Android. Structured logging for Cloud Logging + Monitoring alerts.